

Total No. of Questions : 4]

SEAT No. :

PB-254

[Total No. of Pages : 2

[6270]-42

B.E. (Civil) (Insem.)

HYDROPOWER ENGINEERING

(2019 Pattern) (Semester - VIII) (401013-E) (Elective - V)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the Candidates:

- 1) Solve Q1 or Q2, Q3 or Q4.
- 2) Figures to the right side indicate full marks.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Assume suitable data if necessary.
- 5) Use of logarithmic tables, slide rule, Mollier charts, electronic pocket calculator (non-programmable) and steam tables is allowed.

- Q1) a) What are the effects of climate change and green house effects on hydropower generation and its development? [5]
- b) Explain the classification of hydroelectric power plants based on hydraulic features. [5]
- c) Distinguish between thermal power plant and hydropower plant based on site selection, initial cost, fuel cost, operating cost, maintenance cost, spare, cooling water requirement, Transmission cost, distribution cost, pollution and Life of plants. [5]

OR

- Q2) a) What are the criteria for the selection of site for the hydropower plants? [5]
- b) Differentiate between mini and micro hydropower plants with suitable example. [5]
- c) Which are conventional and Nonconventional energy sources available in India? [5]

- Q3) a) Define : [5]
- i) Plant capacity Factor
 - ii) Plant use Factor
 - iii) Connected load Factor
 - iv) Demand Factor
 - v) Diversity Factor

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- b) Write the significance of diversity Factor on the cost of hydropower generation. [5]
- c) When runoff river plant operates as peak load station with weekly load factor of 30%, all its capacity is firm capacity. What must be minimum flow in the river so that the station may serve as the base load station? It is given that rated installation capacity of generator is 10,000 KW. Operating head is 25m, plant efficiency is 90%. Estimate the daily load Factor of plant if the stream Flow is 20 cumec. [5]

OR

- Q4) a) Write short note on tariff for electrical energy and types of tariffs. [5]
- b) Write note on prediction of load with suitable load curve diagram. [5]
- c) A load on hydroelectric plant varies from minimum 15,000 KW to a maximum 60,000 KW. Two turbo generators of capacities 30,000 KW each have been installed. Calculate [5]
- Total installed capacity of plant
 - Plant factor
 - Maximum demand
 - Load Factor
 - Utilization Factor